Balanced Array

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

You are given an array a of n integers.

The array is called **balanced** if every middle element is equal to the sum of its two neighbors.

In other words, for every index i such that $2 \le i \le n - 1$:

$a_i = a_{i-1} + a_{i+1}$

Determine whether the array is balanced.

Input

The first line contains an integer n ($3 \le n \le 10^5$).

The second line contains n integers a_i ($-10^9 \le a_i \le 10^9$).

Output

Print `YES` if the array is balanced. Otherwise, print `NO`.

Examples

standard input	standard output
5 1 2 1 -1 -2	YES
5 1 2 1 -1 -1	NO